

SOPHIA BARBER

University of Washington, Seattle: Graduate Student

Email: sbarber990@outlook.com (personal), sophiab3@uw.edu (academic)

Phone: 425.998.3499

Education

University of Washington, Seattle

B.S. in Applied Mathematics; Mathematics (double major), *Magna Cum Laude*

Graduated June 2025

GPA: 3.98/4.00

M.S. in Applied Mathematics, *In Progress*

Expected Graduation: June 2026

Current GPA: 3.93/4.00

Relevant

Degree Courses (B.S. and M.S.):

Coursework

- Scientific Computing & Data Analysis (including numerical techniques for differential equations, spectral analysis, filtering methods for signal processing, principal component analysis, orthogonal mode decomposition, and image processing/compression)
- Probability, Stochastic Processes, Stochastic Calculus, Stochastic Differential Equations, Jump Diffusions
- Linear Algebra (Theoretical and Numerical), Numerical Analysis
- Complex Analysis, Real Analysis
- Multivariable and Vector Calculus
- Number Theory
- Ordinary Differential Equations, Partial Different Equations, Waves
- Dynamical Systems & Chaos
- Computational Modeling (Continuous and Discrete), including ordinary differential equation models and some stochastic models

Other:

- Philosophy (logic, reasoning, ethics, philosophical issues in law)
- Mathematical Reasoning/Proof Writing

Work

Graduate Teaching Assistant

September 2025 - June 2026

Experience

University of Washington Department of Applied Mathematics

Courses:

AMATH 352: Applied Linear Algebra and Numerical Analysis (Autumn 2025)

AMATH 351: Intro to Differential Equations and Applications (Winter 2026)

AMATH 301: Beginning Scientific Computing (Spring 2026)

Held office hours weekly to assist students with questions about course content and homework problems; graded and provided feedback on homework assignments for 130 students every other week; wrote solutions to homework assignments; helped professors to proctor and grade exams.

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Programming

Python, MATLAB, Java, LaTeX

Skills

Experience with basic supervised Neural Networks, including Fully Connected Neural Networks (FCNs) and Convolutional Neural Networks (CNNs).

Microsoft Office Specialist (MOS) certification for Word, Excel, PowerPoint 2016

Honors

Award of Excellence

University of Washington Department of Mathematics - June 2025

Awarded based on academic record and faculty nominations

Languages

English (Native Speaker)

Norwegian (Advanced Beginner)

Japanese (Intermediate)

Latin (Intermediate)

Key Skills

Clear communicator, strong writing and presentation skills

Takes initiative, strong work ethic, well-organized

Enjoys collaborating and working with others

Strong analytical thinking and problem-solving skills

Memberships

Society for Industrial and Applied Mathematics (SIAM)

American Mathematical Society (AMS)

Other

American/Canadian Dual Citizen

Willing to relocate
